

Bass Guitar Fretboard & Complete Reference

A visual reference for standard four-string bass: E–A–D–G

Purpose. A searchable visual reference for restarting bass: complete fretboard and theory material, 20 physical exercises, 20 musical exercises, ten ready-made 30-minute programs, and a concise improvisation guide.

Reference contents

- Complete fretboard, intervals and key reference
- Scale and arpeggio diagrams
- Bass-clef notation and four-line TAB
- Bassline construction and improvisation guide
- 40 five-minute exercises and ten practice programs

Diagram system

Scale and exercise pages show the same material three ways: fretboard diagram, bass-clef notation, and four-line TAB. Exercises are designed as independent five-minute blocks.

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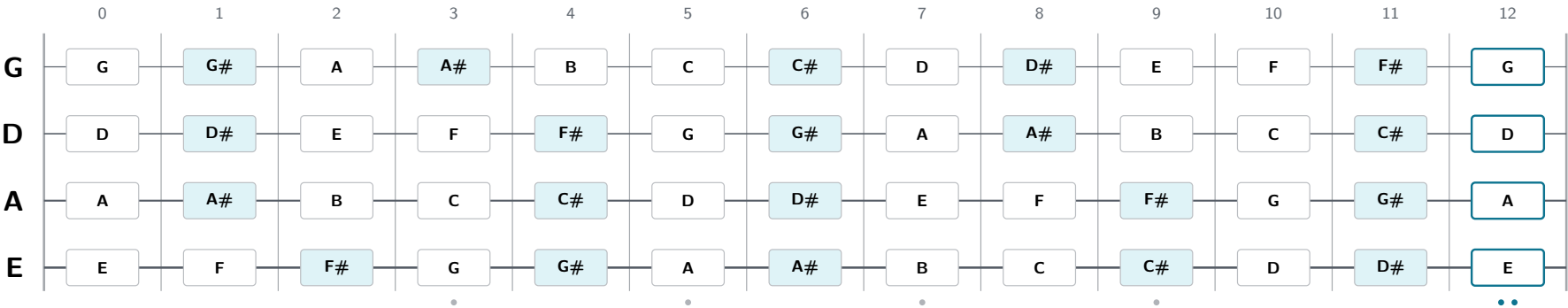
1 Notation conventions

Standard tuning	E–A–D–G from the lowest to the highest string. In TAB, the highest G string is the top line.
Bass clef	Bass guitar is conventionally written one octave above its sounding pitch.
Fret distance	One fret equals one semitone. W denotes a whole step; H denotes a half step.
Scale degrees	Accidentals attached to scale degrees have their conventional meaning: $\flat 3$, $\sharp 4$, $\flat 7$, and so forth.

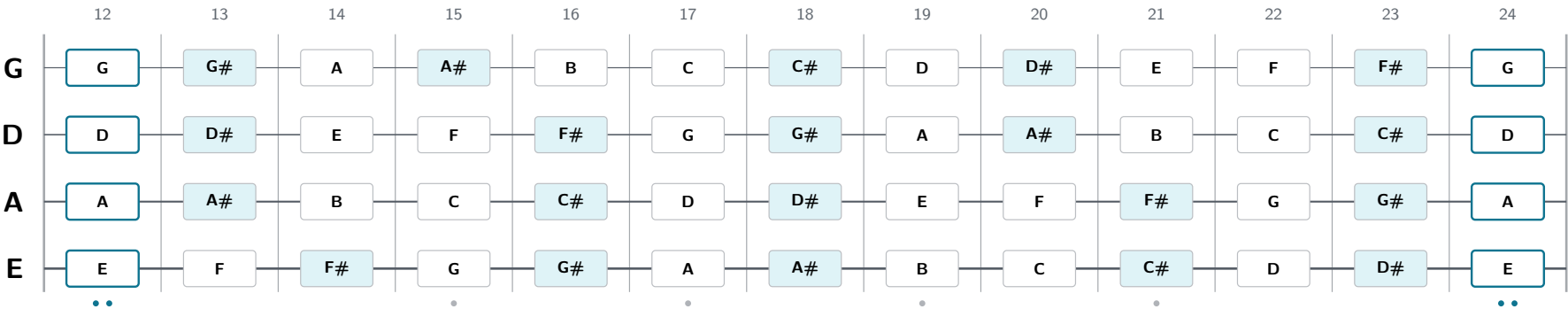
The fretboard diagram, standard notation, and TAB on each scale page describe the same single ascending fingering in D.

2 Complete fretboard: frets 0–24

Accidentals are written with sharps for compactness. Enharmonic spellings still apply: C# = Db, D# = Eb, F# = Gb, G# = Ab, A# = Bb. **Frets 0–12**

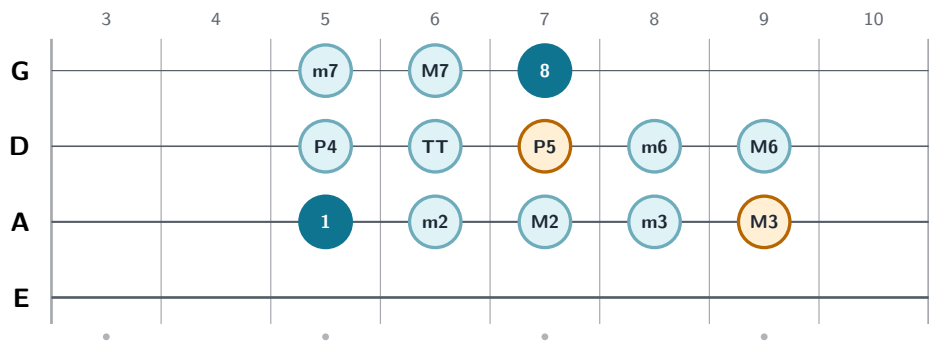


Frets 12–24



3 Interval map on the fretboard

Intervals are the reusable geometry behind scales, arpeggios and basslines. The diagram uses D at A-string fret 5 and shows one nearby occurrence of every interval through the octave.



Interval	Semitones	Degree spelling	Common bass function
Minor second	1	b2	Chromatic approach, Phrygian colour, strongest adjacent tension.
Major second	2	2	Scale connection and suspended colour.
Minor third	3	b3	Defines minor triads and minor-seven chords.
Major third	4	3	Defines major triads and dominant chords.
Perfect fourth	5	4	Strong scale tone; inversion of the perfect fifth.
Tritone	6	#4/b5	Dominant tension and diminished harmony.
Perfect fifth	7	5	Stable support; the most common partner to the root.
Minor sixth	8	b6	Natural-minor colour and approach tone.
Major sixth	9	6	Dorian, major-pentatonic and Latin colour.
Minor seventh	10	b7	Dominant-seven and minor-seven harmony.
Major seventh	11	7	Major-seven colour and leading tone.
Octave	12	8	Same pitch class in a higher register.

4 Reading and rhythm reference

4.1 Bass clef and written range

Bass-clef lines	G–B–D–F–A, from the lowest line upward.
Bass-clef spaces	A–C–E–G, from the lowest space upward.
Open strings	E–A–D–G. Written bass-guitar notation sounds one octave lower than written.
TAB orientation	G is the top line; E is the bottom line. A number is the fret to press.
Accidentals	A sharp or flat applies through the current measure unless cancelled or replaced.
Tie	Joins identical pitches into one sustained duration.
Dot	Adds half of the original note value. A dotted quarter equals three eighth notes.

Rhythmic values on D: whole, halves, quarters and eighths



Value	Length in 4/4	Notes per quarter-note beat	Matching rest
Whole	4 beats	1/4	Whole rest
Half	2 beats	1/2	Half rest
Quarter	1 beat	1	Quarter rest
Eighth	1/2 beat	2	Eighth rest
Sixteenth	1/4 beat	4	Sixteenth rest

4.2 Common articulation and TAB markings

Technique	Common marking	Meaning
Staccato	dot above/below note	Shorten the note while preserving its rhythmic position.
Accent	>	Stronger attack.
Dead / ghost note	x	Muted percussive attack without a clear pitch.
Slide	/ or \ in TAB	Maintain contact while moving to the destination fret.
Hammer-on / pull-off	h / p in TAB	Sound the following note mainly with the fretting hand.
Let ring	tie or “let ring”	Sustain until the indicated release point.

5 Chord symbols and chord construction

Chord type	Example symbol	Formula	Bass priority
Major triad	C	1–3–5	Root first; third states major; fifth supplies stability.
Minor triad	Cm	1–b3–5	Root and minor third define the sound.
Power chord	C5	1–5	Neither major nor minor because the third is absent.
Suspended 2	Csus2	1–2–5	Avoid adding a third unless the harmony resolves.
Suspended 4	Csus4	1–4–5	Fourth replaces the third.
Major seven	Cmaj7	1–3–5–7	Major seventh is the characteristic colour.
Dominant seven	C7	1–3–5–b7	Third and flat seventh form the defining tritone.
Minor seven	Cm7	1–b3–5–b7	Root, flat third and flat seventh identify the chord.
Half-diminished	Cm7b5	1–b3–b5–b7	Flat fifth is essential.
Diminished seven	Cdim7	1–b3–b5–bb7	Symmetrical stack of minor thirds.
Augmented	Caug / C+	1–3–#5	Raised fifth creates the instability.

6 Diatonic harmony

6.1 Major key

I	ii	iii	IV	V	vi	vii°
Major	minor	minor	Major	Major	minor	diminished
maj7	m7	m7	maj7	7	m7	m7b5

6.2 Natural minor key

i	ii°	III	iv	v	VI	VII
minor	diminished	Major	minor	minor	Major	Major
m7	m7b5	maj7	m7	m7	maj7	7

In functional minor harmony the V chord is commonly changed from minor to dominant by raising scale degree 7. This is the practical reason harmonic minor is important.

7 Key signatures and relative minors

Major key	Signature	Relative minor	Main notes altered	Useful bass reference
C	none	A minor	none	All natural notes.
G	1 sharp	E minor	F $\sharp$	G major / E natural minor.
D	2 sharps	B minor	F $\sharp$, C $\sharp$	The example key used throughout the scale atlas.
A	3 sharps	F $\sharp$ minor	F $\sharp$, C $\sharp$, G $\sharp$	Common guitar and bass key.
E	4 sharps	C $\sharp$ minor	F $\sharp$, C $\sharp$, G $\sharp$, D $\sharp$	Open E string is the tonic.
F	1 flat	D minor	B $\flat$	Common horn and Latin key.
B$\flat$	2 flats	G minor	B $\flat$, E $\flat$	Common brass and salsa key.
E$\flat$	3 flats	C minor	B $\flat$, E $\flat$, A $\flat$	Common horn, soul and salsa key.
A$\flat$	4 flats	F minor	B $\flat$, E $\flat$, A $\flat$, D $\flat$	Frequent keyboard and horn key.

Transposition rule: a movable fingering keeps the same interval structure. Shift every note by the same number of frets, then rename the root and scale degrees.

8 Scale index

Scale	Degree formula	Main identifying tone	Typical harmonic fit
Major / Ionian	1 2 3 4 5 6 7	Natural 3 and 7	Major, major 7
Natural minor / Aeolian	1 2 $\flat$ 3 4 5 $\flat$ 6 $\flat$ 7	$\flat$ 6	Minor, minor 7
Dorian	1 2 $\flat$ 3 4 5 6 $\flat$ 7	Natural 6	Minor 7
Phrygian	1 $\flat$ 2 $\flat$ 3 4 5 $\flat$ 6 $\flat$ 7	$\flat$ 2	Minor with lowered-second colour
Lydian	1 2 3 $\sharp$ 4 5 6 7	$\sharp$ 4	Major $\sharp$ 11
Mixolydian	1 2 3 4 5 6 $\flat$ 7	$\flat$ 7	Dominant 7
Locrian	1 $\flat$ 2 $\flat$ 3 4 $\flat$ 5 $\flat$ 6 $\flat$ 7	$\flat$ 5	Half-diminished
Harmonic minor	1 2 $\flat$ 3 4 5 $\flat$ 6 7	Raised 7	Minor key with dominant V7
Melodic minor	1 2 $\flat$ 3 4 5 6 7	Natural 6 and 7	Minor-major 7, modern jazz
Major pentatonic	1 2 3 5 6	Omits 4 and 7	Major harmony
Minor pentatonic	1 $\flat$ 3 4 5 $\flat$ 7	Compact minor set	Minor harmony
Blues	1 $\flat$ 3 4 $\flat$ 5 5 $\flat$ 7	Added $\flat$ 5	Blues vocabulary

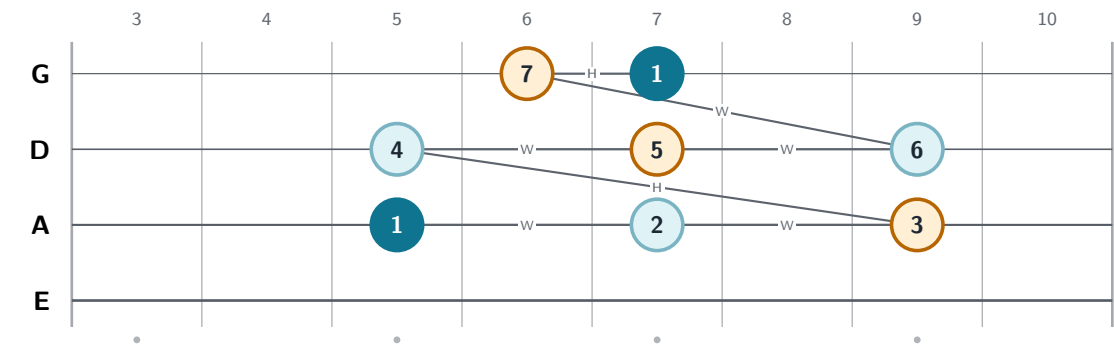
The scale atlas uses D as the example root because A-string fret 5 gives a clear, movable mid-neck position. The degree formula, not the note spelling, is the transposable information.

9 Major (Ionian)

Formula1 2 3 4 5 6 7

Step patternW – W – H – W – W – W – H

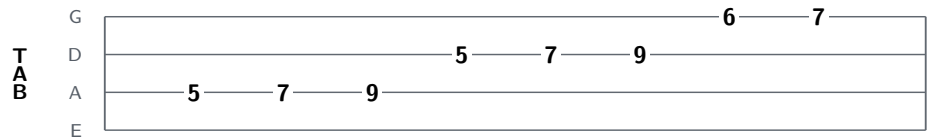
D exampleD–E–F#–G–A–B–C#–D



Standard notation



Bass TAB



Signature soundNatural 3 and natural 7

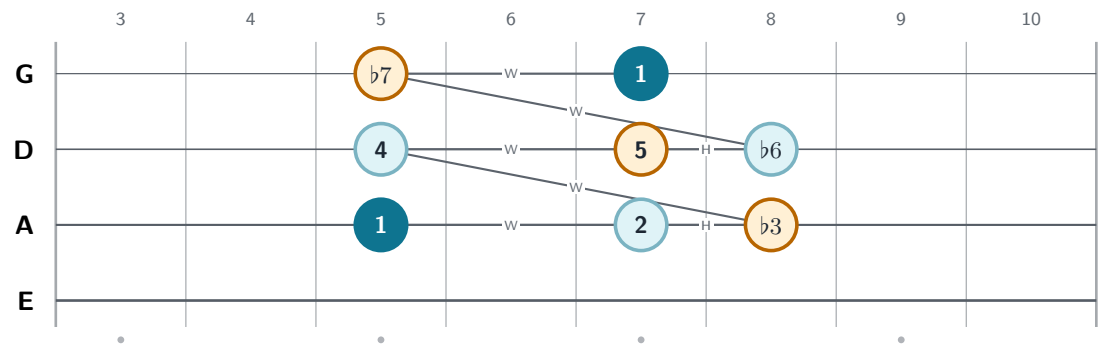
Harmonic fitMajor keys; major and major-7 chords

Overall colourStable, bright, resolved

ComparisonReference scale for all interval formulas.

10 Natural minor (Aeolian)

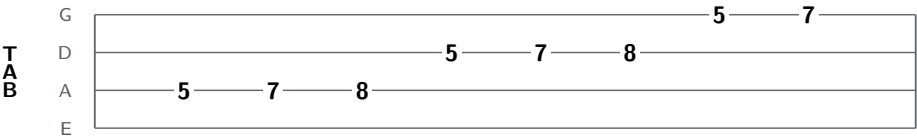
Formula 1 2 $\flat$ 3 4 5 $\flat$ 6 $\flat$ 7
Step pattern W – H – W – W – H – W – W
D example D–E–F–G–A–B $\flat$ –C–D



Standard notation



Bass TAB



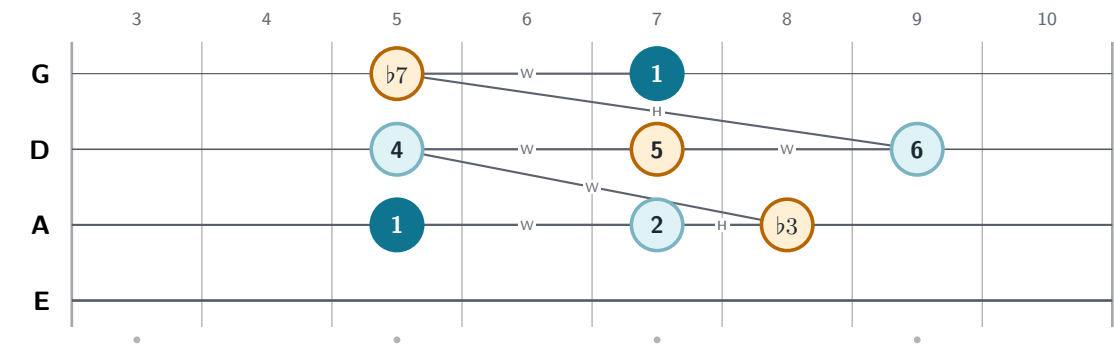
Signature sound $\flat$ 3, $\flat$ 6, $\flat$ 7
Harmonic fit Minor keys; minor and minor-7 chords when the harmony contains $\flat$ 6
Overall colour Dark, settled minor sound
Comparison Relative minor of F major; parallel minor of D major.

11Dorian

Formula1 2 $\flat 3$ 4 5 6 $\flat 7$

Step patternW – H – W – W – W – H – W

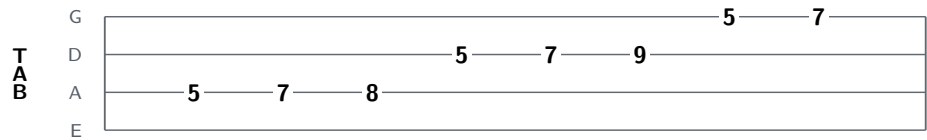
D exampleD–E–F–G–A–B–C–D



Standard notation



Bass TAB



Signature soundNatural 6 over a minor chord

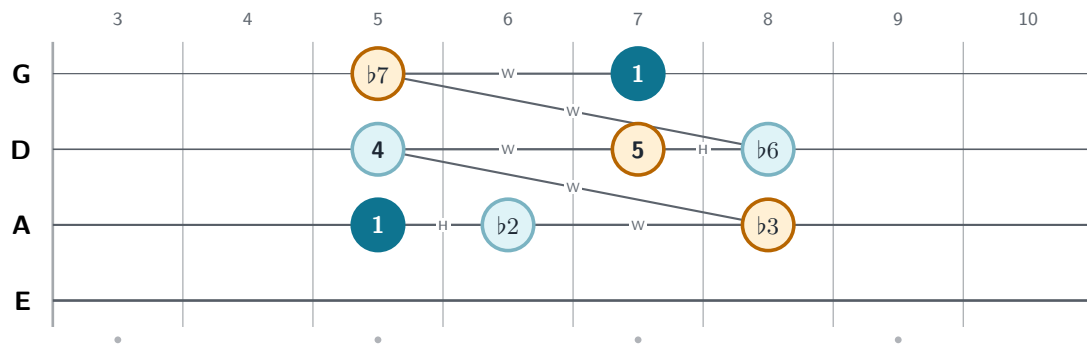
Harmonic fitMinor-7 grooves, funk, modal jazz, Latin harmony

Overall colourMinor with a brighter, more open sixth

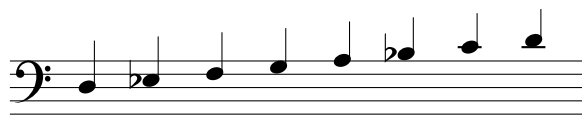
ComparisonNatural minor with $\flat 6$ raised to 6.

12 Phrygian

Formula 1 $\flat$ 2 $\flat$ 3 4 5 $\flat$ 6 $\flat$ 7
Step pattern H – W – W – W – H – W – W
D example D–E $\flat$ –F–G–A–B $\flat$ –C–D



Standard notation



Bass TAB

T A B	G	5 — 7					
	D	5 — 7 — 8					
	A	5 — 6 — 8					
	E						

Signature sound $\flat$ 2
Harmonic fit Minor chords with a strong lowered-second colour; Spanish and metal contexts
Overall colour Dark and compressed around the root
Comparison Natural minor with 2 lowered to $\flat$ 2.

13 Lydian

Formula

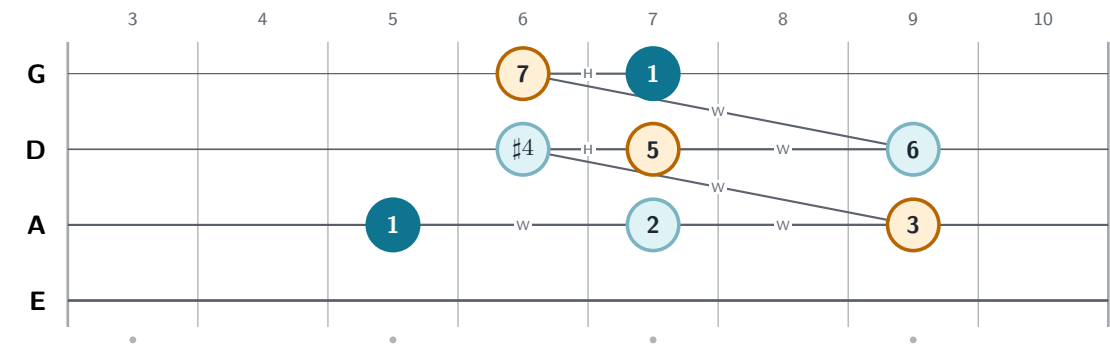
1 2 3 $\sharp 4$ 5 6 7

Step pattern

W – W – W – H – W – W – H

D example

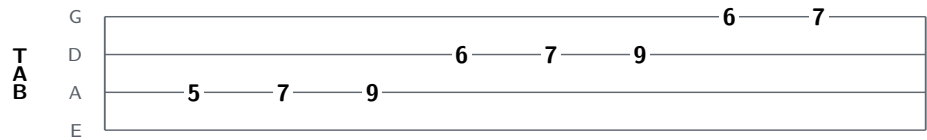
D–E–F $\sharp$ –G $\sharp$ –A–B–C $\sharp$ –D



Standard notation



Bass TAB



Signature sound

$\sharp 4$

Harmonic fit

Major chords with $\sharp 11$; sustained major harmony

Overall colour

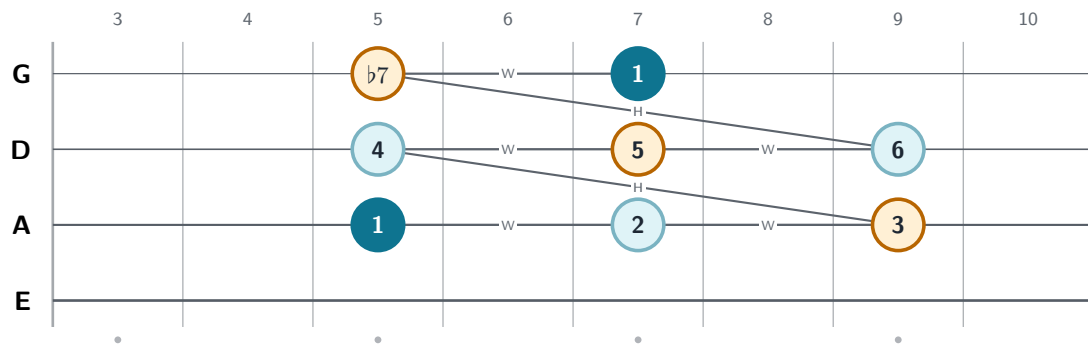
Bright, floating, less final than Ionian

Comparison

Major scale with 4 raised to $\sharp 4$.

14 Mixolydian

Formula 1 2 3 4 5 6 $\flat 7$
Step pattern W – W – H – W – W – H – W
D example D–E–F $\sharp$ –G–A–B–C–D



Standard notation



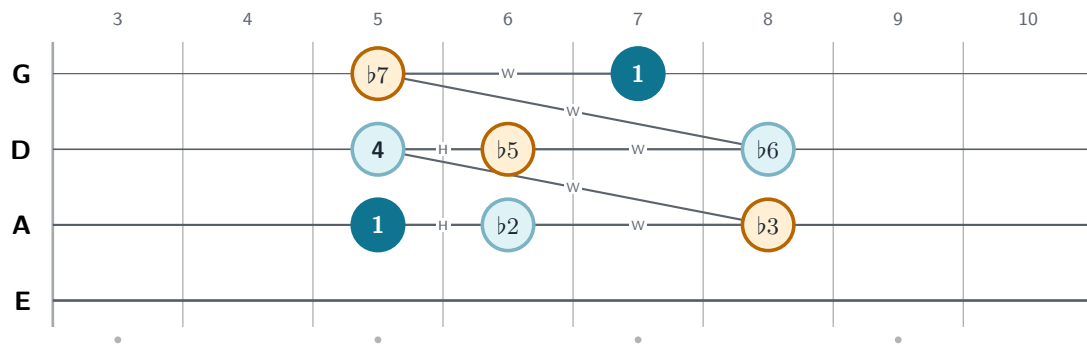
Bass TAB

T A B	G	5 — 7					
	D	5 — 7 — 9					
	A	5 — 7 — 9					
	E						

Signature sound $\flat 7$ over a major third
Harmonic fit Dominant-7 chords, blues, funk, rock and salsa harmony
Overall colour Major with dominant tension
Comparison Major scale with 7 lowered to $\flat 7$.

15 Locrian

Formula 1 $\flat$ 2 $\flat$ 3 4 $\flat$ 5 $\flat$ 6 $\flat$ 7
Step pattern H – W – W – H – W – W – W
D example D–E $\flat$ –F–G–A $\flat$ –B $\flat$ –C–D



Standard notation



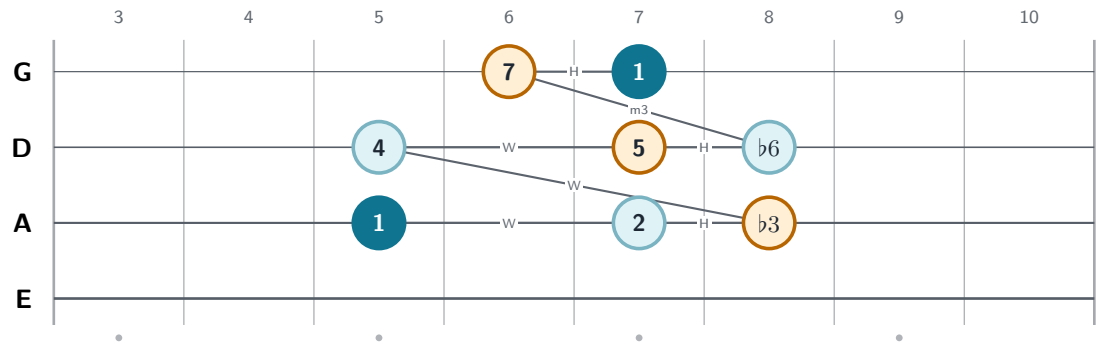
Bass TAB

T A B	G	5 — 7				
	D	5 — 6 — 8				
	A	5 — 6 — 8				
	E					

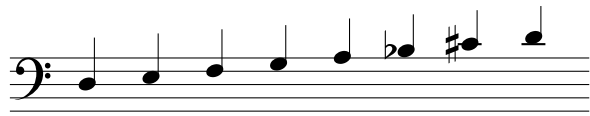
Signature sound $\flat$ 2 and $\flat$ 5
Harmonic fit Half-diminished / minor-7-flat-5 chords
Overall colour Unstable because the perfect fifth is absent
Comparison Natural minor with 2 and 5 lowered.

16 Harmonic minor

Formula	1 2 $\flat 3$ 4 5 $\flat 6$ 7
Step pattern	W – H – W – W – H – A2 – H
D example	D–E–F–G–A–B $\flat$ –C $\sharp$ –D



Standard notation



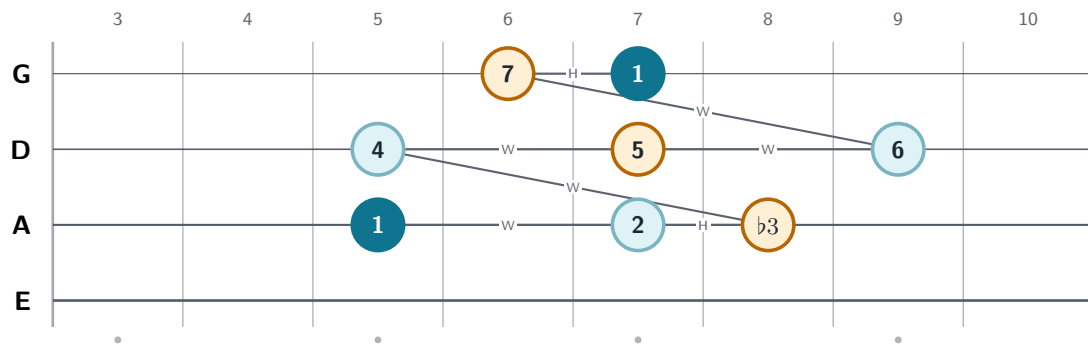
Bass TAB

T A B	G	6 — 7					
	D	5 — 7 — 8					
	A	5 — 7 — 8					
	E						

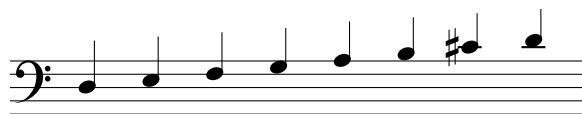
Signature sound	Raised 7 and the augmented second from $\flat 6$ to 7
Harmonic fit	Minor-key cadences; dominant V7 resolving to minor i
Overall colour	Minor with strong leading-tone pull
Comparison	Natural minor with $\flat 7$ raised to 7.

17 Melodic minor

Formula	1 2 $\flat 3$ 4 5 6 7
Step pattern	W – H – W – W – W – W – H
D example	D–E–F–G–A–B–C $\sharp$ –D



Standard notation



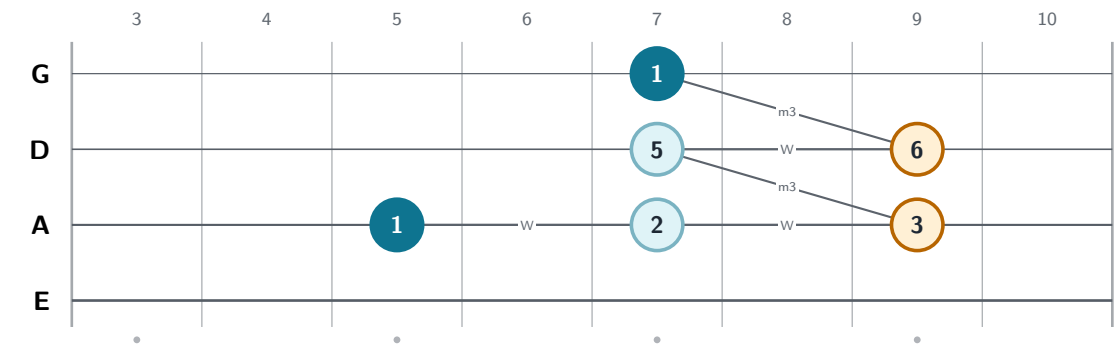
Bass TAB

G	6 — 7
D	5 — 7 — 9
A	5 — 7 — 8
E	

Signature sound	$\flat 3$ with natural 6 and 7
Harmonic fit	Minor-major-7 harmony and modern jazz sounds
Overall colour	Minor tonic with a smooth upper half
Comparison	Diagram uses the ascending/jazz form; classical descent often uses natural minor.

18 Major pentatonic

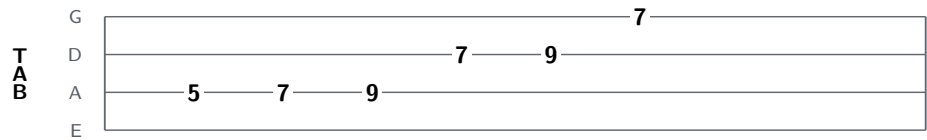
Formula 1 2 3 5 6
Step pattern W – W – m3 – W – m3
D example D–E–F#–A–B–D



Standard notation



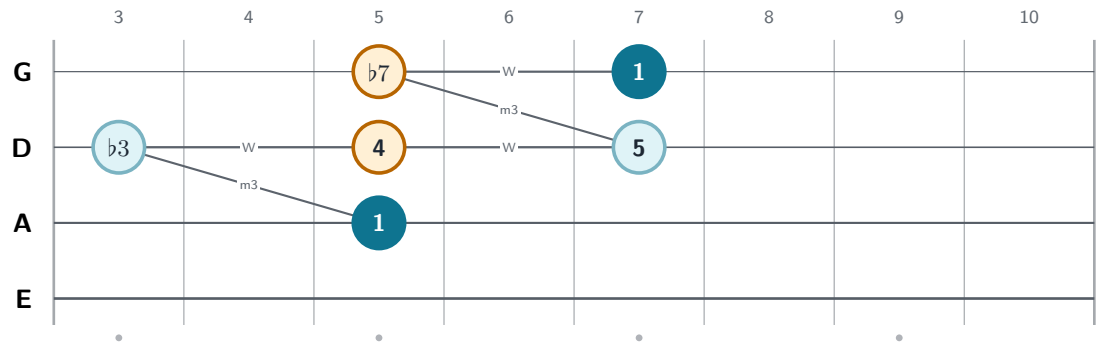
Bass TAB



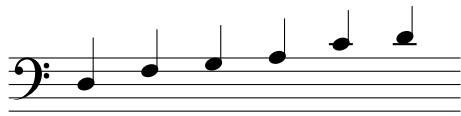
Signature sound No 4 or 7
Harmonic fit Major chords, melodic fills, country, soul and pop
Overall colour Open and consonant
Comparison Major scale with 4 and 7 removed.

19 Minor pentatonic

Formula 1 $\flat$ 3 4 5 $\flat$ 7
Step pattern m3 – W – W – m3 – W
D example D–F–G–A–C–D



Standard notation



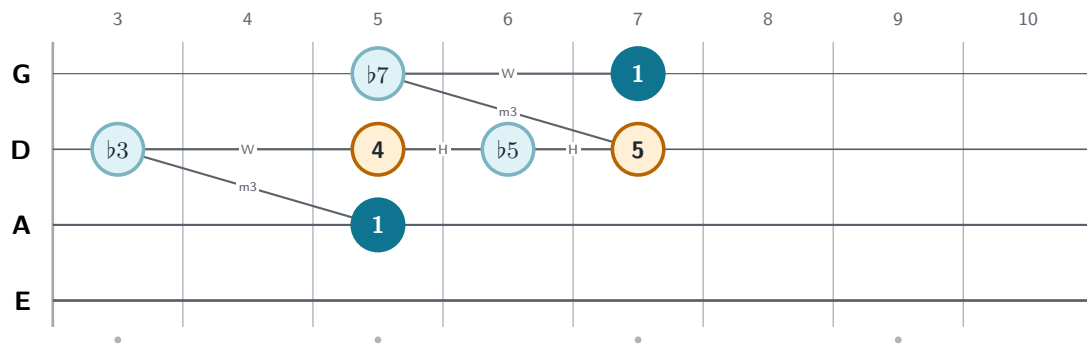
Bass TAB

T A B	G	5 — 7				
	D	3	5	7		
	A	5				
	E					

Signature sound $\flat$ 3 and $\flat$ 7; no 2 or 6
Harmonic fit Minor grooves, blues, rock and funk
Overall colour Strong, compact minor vocabulary
Comparison Natural minor with 2 and $\flat$ 6 removed.

20 Blues scale

Formula 1 $\flat$ 3 4 $\flat$ 5 5 $\flat$ 7
Step pattern m3 – W – H – H – m3 – W
D example D–F–G–A $\flat$ –A–C–D



Standard notation



Bass TAB

TAB	G						5	7
	D	3		5	6	7		
	A	5						
	E							

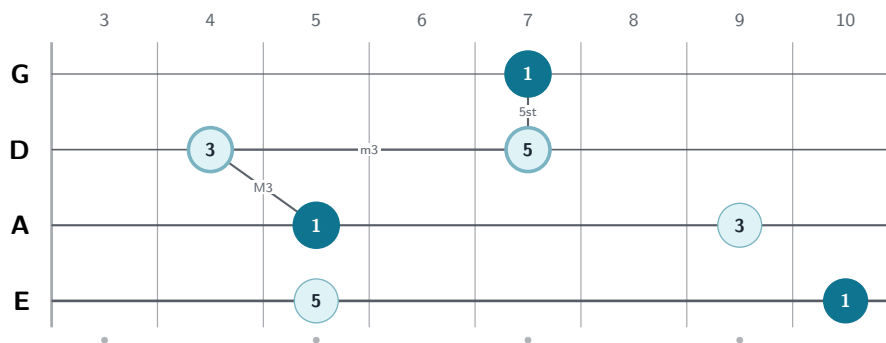
Signature sound $\flat$ 5 passing between 4 and 5
Harmonic fit Blues language and tension inside minor-pentatonic lines
Overall colour Minor pentatonic with an added chromatic pull
Comparison Minor pentatonic plus $\flat$ 5.

21 Chord-tone arpeggio atlas

Scales describe the available pitch collection; arpeggios show the chord itself. In bass lines, roots, thirds, fifths and sevenths usually carry more harmonic information than the remaining scale tones.

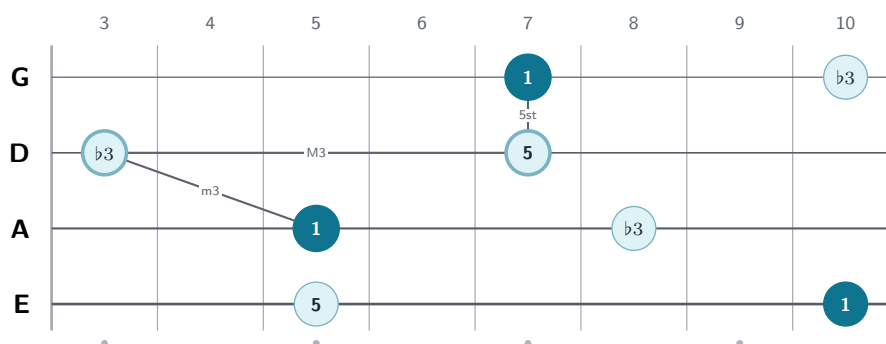
21.1 Major triad

Formula 1 3 5 Example: D–F♯–A–D Use: Major chord skeleton



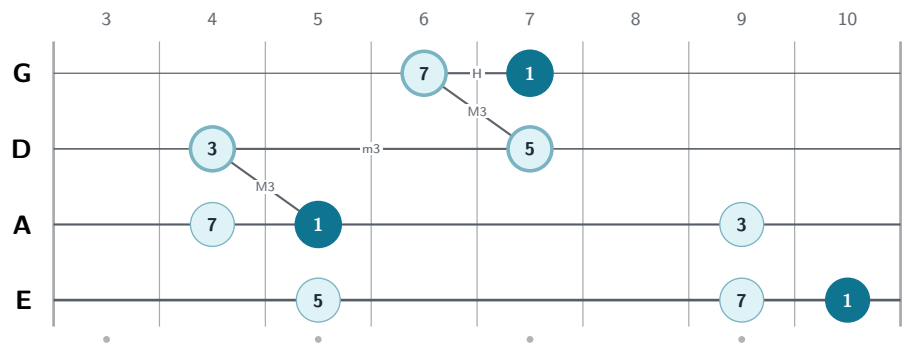
21.2 Minor triad

Formula 1 b3 5 Example: D–F–A–D Use: Minor chord skeleton



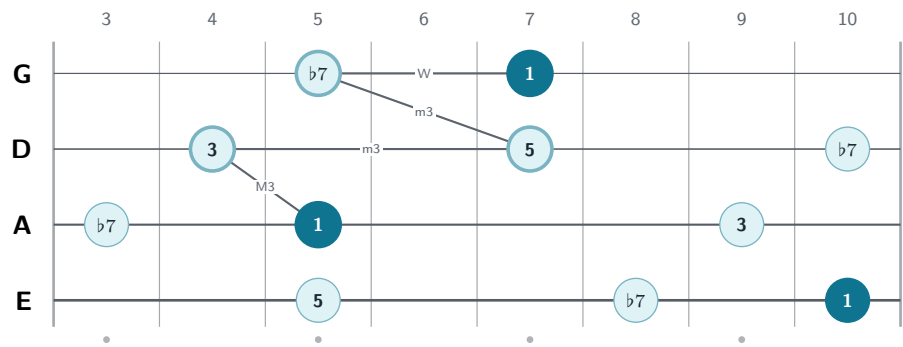
21.3 Major seventh

Formula 1 3 5 7 Example: D-F#-A-C#-D Use: Major-7 harmony



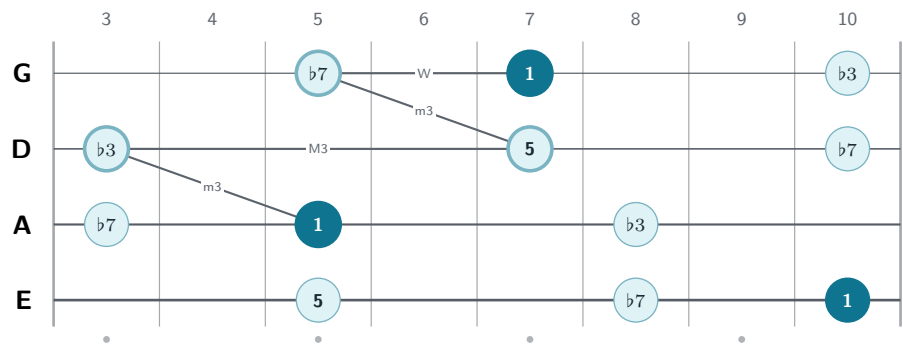
21.4 Dominant seventh

Formula 1 3 5 b7 Example: D-F#-A-C-D Use: Dominant-7 harmony; strong resolution



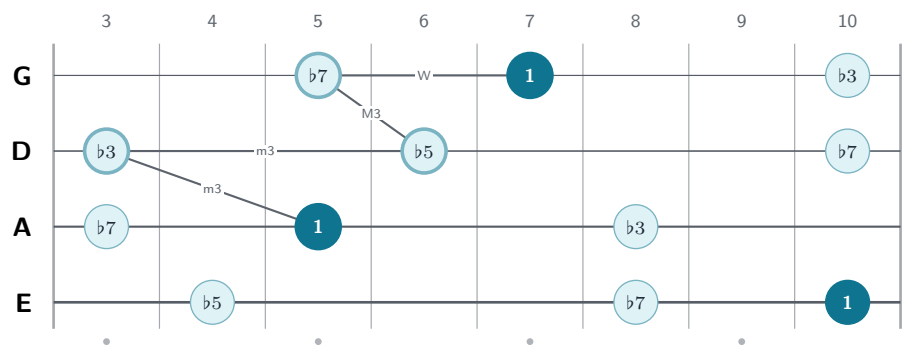
21.5 Minor seventh

Formula 1 b3 5 b7 Example: D-F-A-C-D Use: Minor-7 harmony



21.6 Half-diminished

Formula 1 b3 b5 b7 Example: D-F-A^b-C-D Use: Minor-7-flat-5 harmony



22 How to build a basic bassline

A basic bassline can be built in layers: **rhythm first, roots second, chord tones third, approaches last**. Each added note should clarify the harmony or improve the connection to the next chord.

22.1 Layer 1: map the harmony

For every chord, write its root and chord tones. Example progression:

C	Am	F	G7
C-E-G	A-C-E	F-A-C	G-B-D-F
1-3-5	1-b3-5	1-3-5	1-3-5-b7

22.2 Layer 2: place the roots

Put the root on the strongest structural point, normally beat 1 in a simple 4/4 line:

C	Am	F	G7
C - - -	A - - -	F - - -	G - - -

22.3 Layer 3: add stable chord tones

Fifths are neutral and strong; thirds state major versus minor; sevenths define seventh chords. One quarter-note skeleton is:

C	Am	F	G7
C - G - E - G	A - E - C - E	F - C - A - C	G - D - F - D
1 - 5 - 3 - 5	1 - 5 - b3 - 5	1 - 5 - 3 - 5	1 - 5 - b7 - 5

22.4 Layer 4: connect the next root

On the final subdivision before a chord change, approach the next root from a scale tone or from one fret above/below. The destination note is more important than whether the approach is diatonic or chromatic.

C	Am	F	G7
C - G - E - B	A - E - C - E	F - C - A - F#	G - D - F - B
R - 5 - 3 - approach	R - 5 - b3 - approach	R - 5 - 3 - chromatic	R - 5 - b7 - leading tone
↘ A	↗ F	↗ G	↗ C

22.5 Layer 5: choose rhythm and note length

The same pitch sequence can produce different basslines. Decide whether notes are short, sustained, tied, syncopated or anticipated. Repetition creates identity; small changes at the end of a phrase create direction.

Element	Reference principle
Root	Establishes the chord and tonal center.
Third	States major or minor; use deliberately because it strongly colours the line.
Fifth	Stable support that does not compete with the chord quality.
Seventh	Defines major-7, dominant-7 or minor-7 harmony.
Scale tone	Adds melodic connection while remaining inside the key or chord-scale.
Chromatic approach	One fret above or below a destination chord tone; tension resolves on arrival.
Octave	Reinforces the root with more motion and register contrast.
Rest / sustain	Controls groove as strongly as pitch choice.

Compact formula: choose a rhythm → place the root → add one or two chord tones → approach the next root → repeat the cell with minimal variation.

23 Bassline reference for salsa and Latin harmony

The harmonic construction method does not change: identify the chord, prioritize chord tones, and connect to the next harmony. What changes is the rhythmic placement and interaction with percussion.

Reference point	Application
Repeated cell	A tumbao is usually recognized by a stable repeating rhythmic-harmonic cell rather than by constant melodic variation.
Anticipation	The next root or another chord tone may arrive before the written downbeat and continue across it.
Chord tones	Root and fifth provide stability; third, sixth and seventh identify the exact harmony.
Dominant harmony	Mixolydian and dominant-7 arpeggios expose 3 and $b7$, the defining tritone of a dominant chord.
Minor harmony	Dorian supplies natural 6; natural minor supplies $b6$; use the form present in the actual chord or melody.
Minor-key cadence	Harmonic minor explains the raised 7 and the major-third dominant chord leading back to a minor tonic.
Chromatic connection	A one-fret approach into the anticipated destination is often more useful than running the complete scale.
Space	Sustains and rests leave room for conga, timbales, piano and vocals.

24 Chord-to-scale lookup

Chord / context	First reference	Important degrees	Alternatives / caution
Major triad	Major, major pentatonic	1, 3, 5, 6	Lydian when $\sharp 4$ is supported
Major 7	Major	1, 3, 5, 7	Lydian for $\text{maj}7\sharp 11$
Dominant 7	Mixolydian	1, 3, 5, $b7$	Altered options depend on extensions and resolution
Minor triad	Natural minor or minor pentatonic	1, $b3$, 5	Check whether 6 is natural or flat
Minor 7	Dorian or natural minor	1, $b3$, 5, $b7$	Dorian = 6; Aeolian = $b6$
Minor-major 7	Melodic minor	1, $b3$, 5, 7	Strong tonic-minor colour
Half-diminished	Locrian	1, $b3$, $b5$, $b7$	Locrian natural 2 may appear in some minor-key contexts
Minor V7-i cadence	Harmonic minor around the key center	$b3$, $b6$, 7	The V chord itself is dominant; target its 3 and $b7$
Blues / dominant vamp	Minor pentatonic, blues, Mixolydian	1, $b3/3$, 5, $b7$	Major and minor thirds can coexist stylistically

25 Compact formula sheet

Major	1 2 3 4 5 6 7	Harmonic minor	1 2 $b3$ 4 5 $b6$ 7
Natural minor	1 2 $b3$ 4 5 $b6$ $b7$	Melodic minor	1 2 $b3$ 4 5 6 7
Dorian	1 2 $b3$ 4 5 6 $b7$	Major pentatonic	1 2 3 5 6
Phrygian	1 $b2$ $b3$ 4 5 $b6$ $b7$	Minor pentatonic	1 $b3$ 4 5 $b7$
Lydian	1 2 3 $\sharp 4$ 5 6 7	Blues	1 $b3$ 4 $b5$ 5 $b7$
Mixolydian	1 2 3 4 5 6 $b7$	Major triad	1 3 5
Locrian	1 $b2$ $b3$ 4 $b5$ $b6$ $b7$	Minor triad	1 $b3$ 5
		Dominant 7	1 3 5 $b7$
		Minor 7	1 $b3$ 5 $b7$

26 Complete five-minute exercise library

Every exercise is an independent five-minute block. Each page contains the same three views: the notes on the fretboard, standard bass-clef notation, and matching four-line bass TAB. The physical library develops control of the instrument; the musical library develops hearing, harmony, rhythm and usable bass vocabulary.

- Five-minute method** Spend roughly 30 seconds reading the three representations, four minutes looping the material, and 30 seconds recording the clean tempo and one problem.
- Tempo rule** Begin slowly enough to remain relaxed. Add 4 bpm after three clean repetitions. Accuracy, muting and pulse take priority over speed.
- Transposition** After the written pattern is secure, move it to another root while preserving intervals and rhythm.
- Musical standard** A note is not complete until its start, duration and release are all controlled.

Physical library index

ID	Five-minute block	Page
P1	1-2-3-4 chromatic cell	p. 26
P2	4-3-2-1 reverse cell	p. 27
P3	Permutation 1-3-2-4	p. 28
P4	Permutation 1-4-2-3	p. 29
P5	Same-fret string ladder	p. 30
P6	Diagonal spider outward	p. 31
P7	Diagonal spider inward	p. 32
P8	String-skipping ladder	p. 33
P9	Two-fret position shift	p. 34
P10	Whole-position string shift	p. 35
P11	Movable octave shapes	p. 36
P12	Root-fifth-octave triangle	p. 37
P13	Minor-third and major-third control	p. 38
P14	One-fret diagonal crawl	p. 39
P15	Three-notes-per-string coordination	p. 40
P16	Repeated-note plucking alternation	p. 41
P17	Two-string alternation	p. 42
P18	Wide string-skip control	p. 43
P19	Hammer-on and pull-off cell	p. 44
P20	Four-string endurance pedal	p. 45

Musical library index

ID	Five-minute block	Page
M1	D major scale	p. 46
M2	D natural minor scale	p. 47
M3	D major-pentatonic phrase	p. 48
M4	D minor-pentatonic phrase	p. 49
M5	D blues-scale phrase	p. 50
M6	D major triad	p. 51
M7	D minor triad	p. 52
M8	D dominant-seven arpeggio	p. 53
M9	D minor-seven arpeggio	p. 54
M10	I-IV-V-I root-fifth line	p. 55
M11	I-vi-ii-V root-third targeting	p. 56
M12	ii-V-I seventh-chord chain in C	p. 57

ID	Five-minute block	Page
M13	D-major scale in thirds	p. 58
M14	Chromatic approaches from below	p. 59
M15	Chromatic approaches from above	p. 60
M16	Two-sided enclosures	p. 61
M17	D pedal-tone groove	p. 62
M18	Syncopated D-minor groove	p. 63
M19	Basic Latin two-hit cell	p. 64
M20	Latin ii-V-I-VI turnaround	p. 65

27 Physical technique: 20 five-minute blocks

27.1 P1 - 1-2-3-4 chromatic cell

Five-minute block. Quarter note = 50–100 bpm; eighth notes.

Play: Use fingers 1–2–3–4, then return 4–3–2–1. Repeat on every string.

Fretboard

456789

G

D

A

E

D

D#

E

F

Standard notation

Bass TAB

T

A

B

G

D

A

E

56788765

Target

Progression

Equal attacks and minimum left-hand motion.

Begin at the lower tempo. Add 4 bpm after three clean repetitions; reduce by 6–8 bpm when timing or muting breaks down.

26

27.2 P2 - 4-3-2-1 reverse cell

Five-minute block. Quarter note = 50–100 bpm; eighth notes.

Play: Begin with finger 4 and descend. Keep all unused fingers close to the string.

Fretboard

Standard notation

Bass TAB

G								
D								
A	8	7	6	5	5	6	7	8
E								

Target

Progression

The fourth finger initiates cleanly without collapsing the hand.

Begin at the lower tempo. Add 4 bpm after three clean repetitions; reduce by 6–8 bpm when timing or muting breaks down.

27.3 P3 - Permutation 1-3-2-4

Five-minute block. Quarter note = 45–90 bpm; eighth notes.

Play: Use fingering 1–3–2–4, then 4–2–3–1. Move the pattern to each string.

Fretboard

	4	5	6	7	8	9
G						
D						
A		D	D#	E	F	
E						

Standard notation

A musical staff in bass clef with a 4/4 time signature. It contains a sequence of eighth notes: D (fret 5), D# (fret 6), E (fret 7), F (fret 8), E (fret 7), D# (fret 6), and D (fret 5). The notes are beamed together in pairs: (D, D#), (E, F), (E, D#), and (D, D#).

Bass TAB

G	
D	
A	5 — 7 — 6 — 8 — 8 — 6 — 7 — 5
E	

Target

Progression

No finger moves before its note is required.

Begin at the lower tempo. Add 4 bpm after three clean repetitions; reduce by 6–8 bpm when timing or muting breaks down.

27.4 P4 - Permutation 1-4-2-3

Five-minute block. Quarter note = 40–85 bpm; eighth notes.
Play: Use fingering 1–4–2–3 and reverse. Keep the thumb relaxed behind the neck.

Fretboard

4

5

6

7

8

9

G

D

A

E

D

D#

E

F

Standard notation

Bass TAB

G

D

A

E

T

A

B

5

8

6

7

7

6

8

5

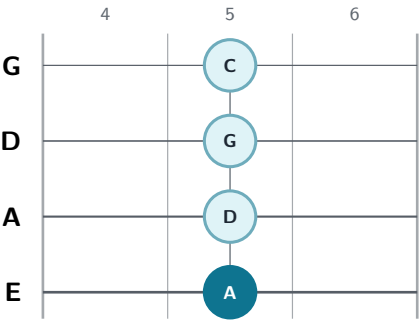
Target
Progression

The 1-to-4 reach stays quiet and controlled.
Begin at the lower tempo. Add 4 bpm after three clean repetitions; reduce by 6–8 bpm when timing or muting breaks down.

27.5 P5 - Same-fret string ladder

Five-minute block. Quarter note = 50–110 bpm; eighth notes.
Play: Cross all four strings at fret 5 and return. Alternate plucking fingers continuously.

Fretboard



Standard notation



Bass TAB

G	5		5
D	5		5
A	5		5
E	5		5

Target

Progression

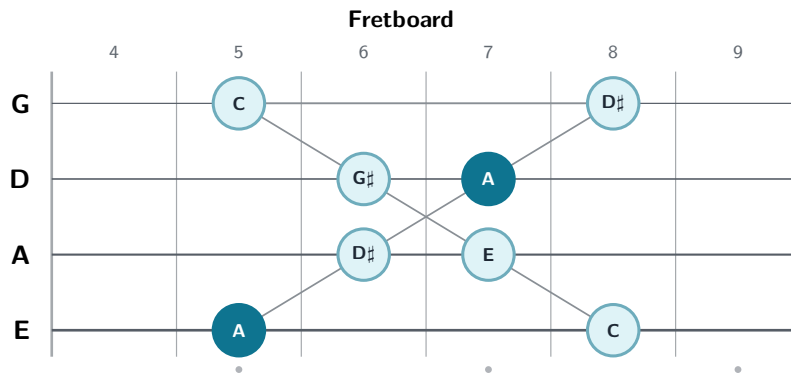
No open-string ringing during either direction change.

Begin at the lower tempo. Add 4 bpm after three clean repetitions; reduce by 6–8 bpm when timing or muting breaks down.

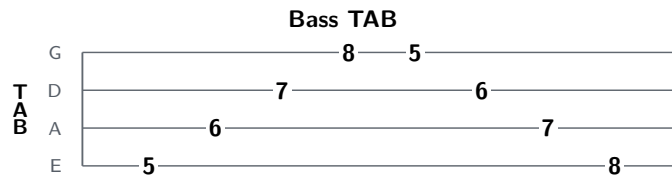
27.6 P6 - Diagonal spider outward

Five-minute block. Quarter note = 40–80 bpm; eighth notes.

Play: Assign fingers 1–2–3–4 to the first diagonal, then 1–2–3–4 to the returning diagonal.



Standard notation



Target

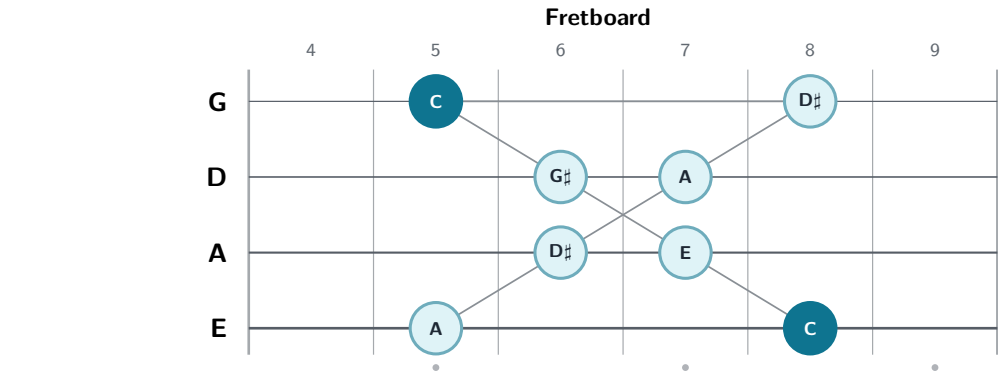
Progression

Every string change is muted before the next note sounds.

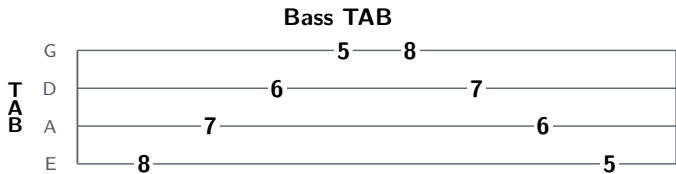
Begin at the lower tempo. Add 4 bpm after three clean repetitions; reduce by 6–8 bpm when timing or muting breaks down.

27.7 P7 - Diagonal spider inward

Five-minute block. Quarter note = 40–80 bpm; eighth notes.
Play: Reverse the diagonal direction from P6 while preserving finger assignments.



Standard notation



Target

Progression

The hand remains centered; the wrist does not twist between strings.
Begin at the lower tempo. Add 4 bpm after three clean repetitions; reduce by 6–8 bpm when timing or muting breaks down.

27.8 P8 - String-skipping ladder

Five-minute block. Quarter note = 45–95 bpm; eighth notes.

Play: Skip one string on every move. Use movable-thumb or floating-thumb muting.

Fretboard

Standard notation

Bass TAB

G	5		6
D	5	6	
A	5	6	
E	5	6	

Target

Progression

Skipped strings remain completely silent.

Begin at the lower tempo. Add 4 bpm after three clean repetitions; reduce by 6–8 bpm when timing or muting breaks down.

27.9 P9 - Two-fret position shift

Five-minute block. Quarter note = 45–90 bpm; eighth notes.
Play: Play frets 5–8, then shift the entire hand to frets 7–10 without pausing.

Fretboard

Standard notation

Bass TAB

G								
D								
A	5	6	7	8	7	8	9	10
E								

Target

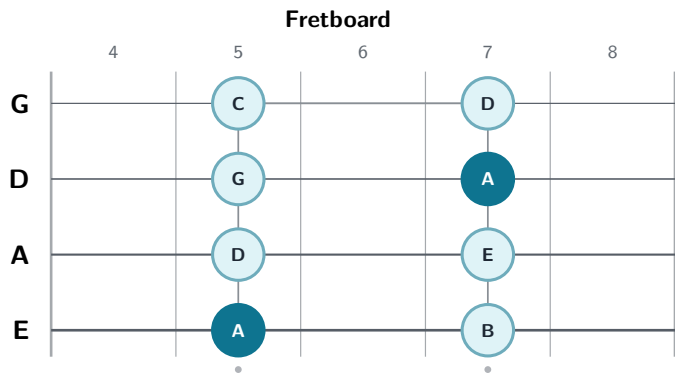
Progression

The shift lands exactly on fret 7 without an audible slide unless deliberately added.

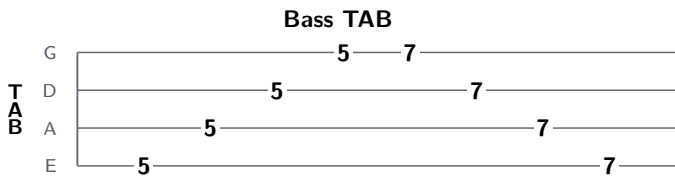
Begin at the lower tempo. Add 4 bpm after three clean repetitions; reduce by 6–8 bpm when timing or muting breaks down.

27.10 P10 - Whole-position string shift

Five-minute block. Quarter note = 45–95 bpm; eighth notes.
Play: Ascend across the strings at fret 5, shift to fret 7, and descend.



Standard notation



Target
Progression

The left hand shifts as one unit; right-hand alternation does not restart.
Begin at the lower tempo. Add 4 bpm after three clean repetitions; reduce by 6–8 bpm when timing or muting breaks down.

27.11 P11 - Movable octave shapes

Five-minute block. Quarter note = 50–100 bpm; eighth notes.
Play: Play each low note followed by its octave two strings higher and two frets forward.

Fretboard

Standard notation

Bass TAB

G	
D	7 — 9 — 10 — 12
A	
E	5 — 7 — 8 — 10

Target

Progression

Both notes of every octave pair match in pitch identity and volume.

Begin at the lower tempo. Add 4 bpm after three clean repetitions; reduce by 6–8 bpm when timing or muting breaks down.

27.12 P12 - Root-fifth-octave triangle

Five-minute block. Quarter note = 55–110 bpm; eighth notes.
Play: Cycle root–fifth–octave–fifth. Move the shape to G and A after one minute each.

Fretboard

Standard notation

Bass TAB

Target

Progression

String crossings sound connected without notes overlapping.

Begin at the lower tempo. Add 4 bpm after three clean repetitions; reduce by 6–8 bpm when timing or muting breaks down.

27.13 P13 - Minor-third and major-third control

Five-minute block. Quarter note = 50–95 bpm; eighth notes.
Play: Alternate the minor third and major third above D. Use fingers that keep the hand relaxed.

Fretboard

A fretboard diagram for the D string (4th line). Frets 2 through 6 are labeled at the top. The D string is labeled on the left. Three notes are circled: F (3rd fret, light blue), F# (4th fret, light blue), and D (5th fret, dark blue). A line connects the F and F# notes. A dot is placed below the 3rd fret line, and another dot is placed below the 5th fret line.

Standard notation

A musical staff in bass clef with a 4/4 time signature. The melody consists of eighth notes: D (3rd fret), E (4th fret), F (5th fret), F# (6th fret), E (4th fret), D (3rd fret), C# (2nd fret), B (1st fret), A (open), G (open), F (3rd fret), E (2nd fret), D (1st fret), C (open), B (open), A (open), G (open).

Bass TAB

G										
D	3 — 4 — 3 — 4 — 3 —									
A	5 — 5 — 5 —									
E										

Target

Progression

The one-fret distinction is visually, physically and aurally clear.

Begin at the lower tempo. Add 4 bpm after three clean repetitions; reduce by 6–8 bpm when timing or muting breaks down.

27.14 P14 - One-fret diagonal crawl

Five-minute block. Quarter note = 40–85 bpm; eighth notes.

Play: Move one fret higher with each higher string, then reverse by one fret per string.

Fretboard

Standard notation

Bass TAB

	G	D	A	E
T				
A				
B				

G: 7 — 6
 D: 6 — 5
 A: 5 — 4
 E: 4 — 3

Target

Progression

Finger placement stays immediately behind each fret.

Begin at the lower tempo. Add 4 bpm after three clean repetitions; reduce by 6–8 bpm when timing or muting breaks down.

27.15 P15 - Three-notes-per-string coordination

Five-minute block. Quarter note = 50–105 bpm; eighth notes.
Play: Use the D-major fingering but treat it as a coordination drill. Reverse after the octave.

Fretboard

Standard notation

Bass TAB

G						6	7
D	5					7	9
A	5	7	9				
E							

Target

Progression

String changes do not disturb the eighth-note pulse.

Begin at the lower tempo. Add 4 bpm after three clean repetitions; reduce by 6–8 bpm when timing or muting breaks down.

27.16 P16 - Repeated-note plucking alternation

Five-minute block. Quarter note = 60–140 bpm; eighth notes.
Play: Play one D repeatedly with strict index-middle alternation. Start the next repetition with the opposite finger.

Fretboard

G

D

A

E

4

5

6

D

Standard notation

4

Bass TAB

T

A

B

G

D

A

E

5

5

5

5

5

5

5

5

Target

Progression

No change in tone, volume or timing between plucking fingers.

Begin at the lower tempo. Add 4 bpm after three clean repetitions; reduce by 6–8 bpm when timing or muting breaks down.

27.17 P17 - Two-string alternation

Five-minute block. Quarter note = 55–125 bpm; eighth notes.
Play: Alternate A and D strings while maintaining index-middle alternation.

Fretboard

456

G

D

A

E

G

D

Standard notation

Bass TAB

T

A

B

E

5

5

5

5

5

5

5

5

Target

Progression

The lower string is muted immediately after each change.

Begin at the lower tempo. Add 4 bpm after three clean repetitions; reduce by 6–8 bpm when timing or muting breaks down.

27.18 P18 - Wide string-skip control

Five-minute block. Quarter note = 40–90 bpm; eighth notes.

Play: Alternate between E and G strings. Keep A and D muted throughout.

Fretboard

Standard notation

Bass TAB

G	5 — 7 — 8 — 10			
D				
A				
E	5 — 7 — 8 — 10			

Target

The attack remains centered despite the two-string leap.

Progression

Begin at the lower tempo. Add 4 bpm after three clean repetitions; reduce by 6–8 bpm when timing or muting breaks down.

Five-minute block. Quarter note = 45–90 bpm; eighth notes.
Play: Pluck the first note of each four-note group; use hammer-ons and pull-offs for the remaining notes.

Play: Pluck the first note of each four-note group; use hammer-ons and pull-offs for the remaining notes.

Standard notation

Legato notes match the plucked note in volume and rhythmic placement.
Begin at the lower tempo. Add 4 bpm after three clean repetitions; reduce by 6–8 bpm when timing or muting breaks down.

Five-minute block. Quarter note = 70–140 bpm; eighth notes.
Play: Play four repeated notes on each string without stopping. Reverse the string order on alternate repetitions.

Diagram illustrating a DNA double helix structure with four base pairs. The bases are labeled G, D, A, and E on the left strand, and D, A, E, and B on the right strand. The bases are connected by vertical lines, forming a ladder-like structure. The bases are numbered 6, 7, and 8 above the right strand. A dot is present at the bottom center of the diagram.

[illegible]

TAB

Maintain relaxed hands and consistent sound for the full five minutes.

Begin at the lower tempo. Add 4 bpm after three clean repetitions; reduce by 6–8 bpm when timing or muting breaks down.

28 Musical control: 20 five-minute blocks

28.1 M1 - D major scale

Five-minute block. Quarter note = 60–120 bpm; eighth notes.
Play: Play ascending, then reverse. Say either note names or degrees 1–2–3–4–5–6–7–8.

Fretboard

Standard notation

Bass TAB

G						6	7
D	5					7	9
A	5	7	9				
E							

Target

Progression

Hear the half steps F-sharp–G and C-sharp–D.

Begin at the lower tempo. Add 4 bpm after three clean repetitions; reduce by 6–8 bpm when timing or muting breaks down.

28.2 M2 - D natural minor scale

Five-minute block. Quarter note = 55–110 bpm; eighth notes.
Play: Play ascending and descending. Compare F and B-flat with the major-scale F-sharp and B.

Fretboard

Standard notation

Bass TAB

Target

Progression

The flat third, flat sixth and flat seventh sound intentional.

Begin at the lower tempo. Add 4 bpm after three clean repetitions; reduce by 6–8 bpm when timing or muting breaks down.

28.3 M3 - D major-pentatonic phrase

Five-minute block. Quarter note = 65–125 bpm; eighth notes.

Play: Loop the phrase over a D-major backing track. Change only the rhythm after the notes feel secure.

Fretboard

Standard notation

Bass TAB

G	7				
D	7 — 9 — 9 — 7				
A	5 — 7 — 9				
E					

Target

Progression

Every note sounds stable against a D-major chord.

Begin at the lower tempo. Add 4 bpm after three clean repetitions; reduce by 6–8 bpm when timing or muting breaks down.

28.4 M4 - D minor-pentatonic phrase

Five-minute block. Quarter note = 60–120 bpm; eighth notes.
Play: Loop over Dm7 or a D-minor groove. End alternate repetitions on F, then D.

Fretboard

Standard notation

Bass TAB

G				5	—	7	—	5	
D			5	—	7				7
A	5	—	8						
E									

Target

Progression

The phrase resolves differently on the flat third and root.

Begin at the lower tempo. Add 4 bpm after three clean repetitions; reduce by 6–8 bpm when timing or muting breaks down.

28.5 M5 - D blues-scale phrase

Five-minute block. Quarter note = 55–110 bpm; eighth notes.
Play: Treat A-flat as a passing tension between G and A rather than a resting point.

Fretboard

Standard notation

Bass TAB

G					5	7	5
D		5	6	7			
A	5	8					
E							

Target The blue note resolves immediately and does not obscure the chord.

Progression Begin at the lower tempo. Add 4 bpm after three clean repetitions; reduce by 6–8 bpm when timing or muting breaks down.

28.6 M6 - D major triad

Five-minute block. Quarter note = 60–115 bpm; eighth notes.
Play: Sing or say 1–3–5–8 while playing. Move the same shape to G and A.

Fretboard

G
D
A
E

3 4 5 6 7 8

F# D A

Standard notation

Bass TAB

G	7									
D	4		7		7		4		7	
A	5					5				
E										

Target
Progression

The major third is audible as the defining chord tone.
Begin at the lower tempo. Add 4 bpm after three clean repetitions; reduce by 6–8 bpm when timing or muting breaks down.

28.7 M7 - D minor triad

Five-minute block. Quarter note = 60–115 bpm; eighth notes.

Play: Say 1-flat3-5-8. Alternate one repetition of M6 with one repetition of M7.

Fretboard

2 3 4 5 6 7 8

G D A E

Standard notation

Bass TAB

G 7

D 3 7 7 3 7

A 5 5

E 2

Target

Progression

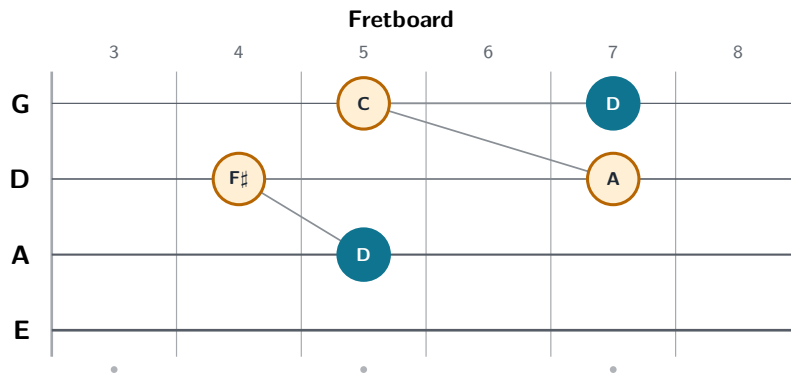
The ear recognizes the one-fret major/minor change immediately.

Begin at the lower tempo. Add 4 bpm after three clean repetitions; reduce by 6–8 bpm when timing or muting breaks down.

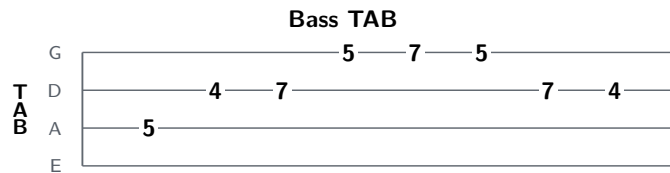
28.8 M8 - D dominant-seven arpeggio

Five-minute block. Quarter note = 55–110 bpm; eighth notes.

Play: Play over D7. Emphasize F-sharp and C, the notes that define dominant quality.



Standard notation



Target

Progression

The flat seventh sounds like controlled tension, not a wrong major seventh.

Begin at the lower tempo. Add 4 bpm after three clean repetitions; reduce by 6–8 bpm when timing or muting breaks down.

28.9 M9 - D minor-seven arpeggio

Five-minute block. Quarter note = 55–110 bpm; eighth notes.
Play: Play over Dm7. Compare directly with M8.

Fretboard

Standard notation

Bass TAB

G	5 — 7 — 5				
D	3 — 7		7 — 3		
A	5				
E					

Target

Progression

Flat third and flat seventh clearly establish minor-seven harmony.

Begin at the lower tempo. Add 4 bpm after three clean repetitions; reduce by 6–8 bpm when timing or muting breaks down.

28.10 M10 - I-IV-V-I root-fifth line

Five-minute block. Quarter note = 70–125 bpm; quarter notes.
Play: Two notes per chord: D–A, G–D, A–E, D–A. Use a D-major I–IV–V–I backing track.

Fretboard

45678

G

D

A

E

G

D

A

E

Standard notation



Bass TAB

G

D

A

E

T

A

B

7577

5575

Target

Progression

Every chord change is audible from the destination root.

Begin at the lower tempo. Add 4 bpm after three clean repetitions; reduce by 6–8 bpm when timing or muting breaks down.

55

28.11 M11 - I-vi-ii-V root-third targeting

Five-minute block. Quarter note = 60–105 bpm; quarter notes.
Play: Play D–F-sharp, B–D, E–G, A–C-sharp over D–Bm–Em–A.

Fretboard

	3	4	5	6	7	8
G						
D		F#	G			
A		C#	D		E	
E			A		B	

Standard notation

Bass TAB

G							
D		4			5		
A	5		5	7		4	
E			7			5	

Target
Progression

The third reveals each chord quality while the line remains compact.
Begin at the lower tempo. Add 4 bpm after three clean repetitions; reduce by 6–8 bpm when timing or muting breaks down.

28.12 M12 - ii-V-I seventh-chord chain in C

Five-minute block. Quarter note = 55–100 bpm; eighth notes.

Play: Arpeggiate Dm7, G7 and Cmaj7. Listen for F remaining common between Dm7 and G7, then resolving to E over Cmaj7.

Fretboard

Standard notation

Bass TAB

G	5 ————— 7 — 10 ————— 5									
D	3 — 7 ————— 5 — 9 ————— 5 — 9 ————— 5									
A	5 ————— 3 — 7 ————— 3 — 7 —————									
E										

Target

Chord tones track the harmony without relying on a single scale.

Progression

Begin at the lower tempo. Add 4 bpm after three clean repetitions; reduce by 6–8 bpm when timing or muting breaks down.

28.13 M13 - D-major scale in thirds

Five-minute block. Quarter note = 45–90 bpm; eighth notes.
Play: Play adjacent diatonic thirds: 1–3, 2–4, 3–5 and onward.

Fretboard

Standard notation

Bass TAB

G																6	7	6	9	7	9
D	5					7	5	9	7	9											
A	5	9	7	9																	
E																					

Target

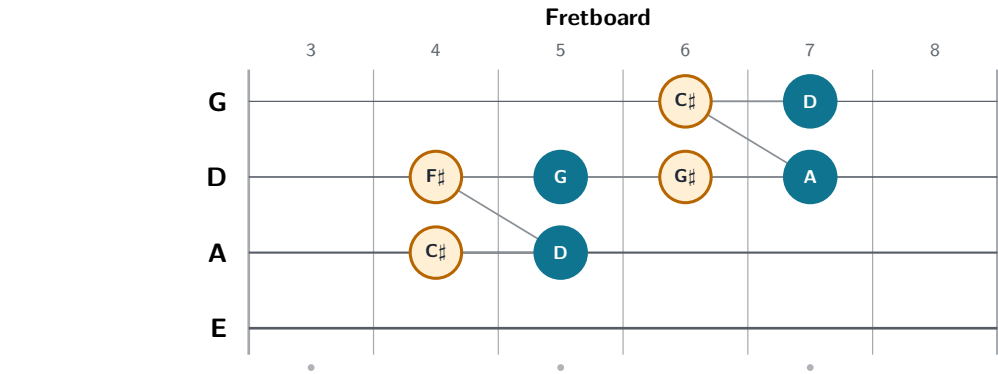
Progression

See the scale as relationships rather than a straight run.

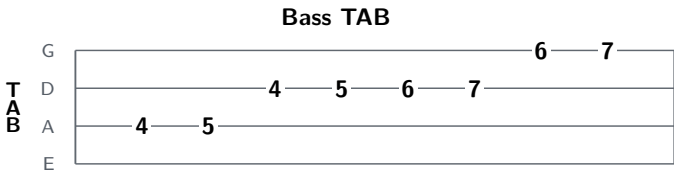
Begin at the lower tempo. Add 4 bpm after three clean repetitions; reduce by 6–8 bpm when timing or muting breaks down.

28.14 M14 - Chromatic approaches from below

Five-minute block. Quarter note = 55–105 bpm; eighth notes.
Play: Approach D, G, A and high D from one fret below. Place each target on a strong beat.



Standard notation

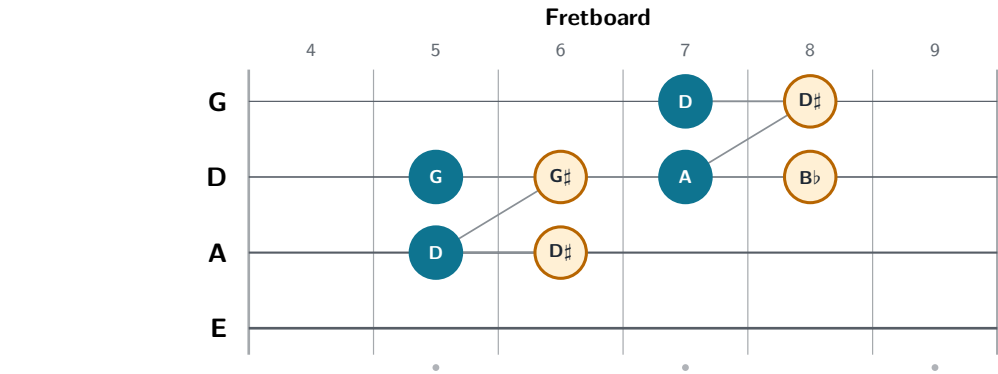


Target
Progression

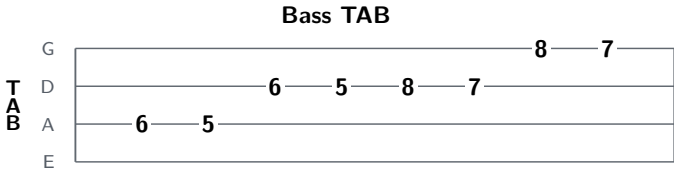
The chromatic note points clearly toward the following chord tone.
Begin at the lower tempo. Add 4 bpm after three clean repetitions; reduce by 6–8 bpm when timing or muting breaks down.

28.15 M15 - Chromatic approaches from above

Five-minute block. Quarter note = 55–105 bpm; eighth notes.
Play: Approach each target from one fret above. Keep the approach short and the target longer when improvising.



Standard notation



Target
Progression

Tension consistently resolves downward by a semitone.
Begin at the lower tempo. Add 4 bpm after three clean repetitions; reduce by 6–8 bpm when timing or muting breaks down.

28.16 M16 - Two-sided enclosures

Five-minute block. Quarter note = 45–90 bpm; eighth notes.

Play: Surround each target with one scale step above and one chromatic step below: E–C-sharp–D, A–F-sharp–G, and so on.

Fretboard

Standard notation

28.17 M17 - D pedal-tone groove

Five-minute block. Quarter note = 70–125 bpm; eighth notes.
Play: Return repeatedly to D while changing the upper note. Accent beats 1 and 3, then move the accents to 2 and 4.

Fretboard

Standard notation

Bass TAB

G	6 — 7				
D	7	4	7		
A	5	5	5		
E					

Target

Progression

The pedal root anchors the phrase while upper tones create motion.

Begin at the lower tempo. Add 4 bpm after three clean repetitions; reduce by 6–8 bpm when timing or muting breaks down.

28.18 M18 - Syncopated D-minor groove

Five-minute block. Quarter note = 65–110 bpm; eighth-note grid.
Play: Count 1-and-2-and-3-and-4-and. Respect every written rest and let silence shape the line.

Fretboard

Diagram of a bass fretboard with strings G, D, A, E. Frets 4 through 9 are marked. Notes are placed as follows: C (orange circle) on G string at fret 5; D (blue circle) on G string at fret 7; A (orange circle) on D string at fret 7; D (blue circle) on A string at fret 5; F (orange circle) on A string at fret 8.

Standard notation

Standard notation in 4/4 time. The bass line consists of eighth notes and rests. The notes are C, D, A, D, F, corresponding to the fretboard diagram. The rhythm is 1-and-2-and-3-and-4-and.

Bass TAB

Bass TAB notation for the syncopated D-minor groove. The strings are G, D, A, E. The fret numbers are 5, 7, 7, 5, 8.

Target Entrances remain accurate after rests; notes do not bleed into silence.

Progression Begin at the lower tempo. Add 4 bpm after three clean repetitions; reduce by 6–8 bpm when timing or muting breaks down.

28.19 M19 - Basic Latin two-hit cell

Five-minute block. Quarter note = 70–110 bpm; eighth-note grid.
Play: Over Dm7 then G7, play the fifth on 2-and and the root on beat 4. Do not add notes on beats 1 or 3.

Fretboard

Standard notation

Bass TAB

Target

Progression

The offbeat arrives without rushing and beat 4 points into the next measure.

Begin at the lower tempo. Add 4 bpm after three clean repetitions; reduce by 6–8 bpm when timing or muting breaks down.

28.20 M20 - Latin ii-V-I-VI turnaround

Five-minute block. Quarter note = 70–105 bpm; eighth-note grid.

Play: Four bars: Dm7–G7–Cmaj7–A7. In each bar place the fifth on 2-and and the root on beat 4.

Fretboard

Diagram of a bass fretboard from fret 2 to 8. Notes are indicated by circles: D (7th fret, G string), A (7th fret, D string), E (6th fret, A string), and G (5th fret, E string). The notes represent the 5th and root of the chords Dm7, G7, Cmaj7, and A7.

Standard notation - line 1

Standard notation for line 1 of the bass line, showing a sequence of eighth notes in 4/4 time.

Standard notation - line 2

Standard notation for line 2 of the bass line, showing a sequence of eighth notes in 4/4 time.

Bass TAB

Bass TAB notation for the two lines, showing fret numbers (7, 5, 5, 3, 7, 5) for each string (G, D, A, E).

Target

Progression

The beat-4 root anticipates and propels the next chord while the clave remains undisturbed.

Begin at the lower tempo. Add 4 bpm after three clean repetitions; reduce by 6–8 bpm when timing or muting breaks down.

29 Ten ready-made 30-minute programs

Choose one program and follow it without redesigning the session. Each row is exactly five minutes. On a difficult day, keep the listed order but lower the tempo; on an easy day, transpose the final two musical blocks.

Program 1 - clean restart

Time	Block	Purpose
0–5	P1 (p. 26)	Basic left-hand coordination
5–10	P5 (p. 30)	String crossing and muting
10–15	P16 (p. 41)	Right-hand alternation
15–20	M1 (p. 46)	Major-scale map
20–25	M6 (p. 51)	Major chord tones
25–30	M10 (p. 55)	Simple progression

Program 2 - fretting-hand control

Time	Block	Purpose
0–5	P3 (p. 28)	Permutation
5–10	P4 (p. 29)	Wide finger order
10–15	P9 (p. 34)	Position shift
15–20	P13 (p. 38)	Third control
20–25	M1 (p. 46)	Scale application
25–30	M14 (p. 59)	Target-note approach

Program 3 - right hand and muting

Time	Block	Purpose
0–5	P5 (p. 30)	Four-string crossing
5–10	P8 (p. 33)	String skips
10–15	P16 (p. 41)	Alternating fingers
15–20	P17 (p. 42)	Two-string alternation
20–25	P20 (p. 45)	Endurance
25–30	M18 (p. 63)	Rests and releases

Program 4 - shifting and register

Time	Block	Purpose
0–5	P9 (p. 34)	Two-fret shift
5–10	P10 (p. 35)	Whole-position shift
10–15	P11 (p. 36)	Octave movement
15–20	P14 (p. 39)	Diagonal position control
20–25	M13 (p. 58)	Scale in thirds
25–30	M17 (p. 62)	Register around a pedal

Program 5 - major-key foundation

Time	Block	Purpose
0–5	P15 (p. 40)	Major fingering coordination
5–10	M1 (p. 46)	D major
10–15	M3 (p. 48)	Major pentatonic
15–20	M6 (p. 51)	Major triad
20–25	M10 (p. 55)	I–IV–V–I
25–30	M11 (p. 56)	I–vi–ii–V

Program 6 - minor-key foundation

Time	Block	Purpose
0–5	P13 (p. 38)	Minor/major third distinction
5–10	M2 (p. 47)	Natural minor
10–15	M4 (p. 49)	Minor pentatonic
15–20	M7 (p. 52)	Minor triad
20–25	M9 (p. 54)	Minor-seven arpeggio
25–30	M18 (p. 63)	Minor groove

Program 7 - chord tones and voice leading

Time	Block	Purpose
0–5	P12 (p. 37)	Root/fifth/octave
5–10	M6 (p. 51)	Major triad
10–15	M7 (p. 52)	Minor triad
15–20	M8 (p. 53)	Dominant seven
20–25	M9 (p. 54)	Minor seven
25–30	M12 (p. 57)	ii–V–I chain

Program 8 - groove and syncopation

Time	Block	Purpose
0–5	P16 (p. 41)	Even pulse
5–10	P17 (p. 42)	String-change pulse
10–15	M17 (p. 62)	Pedal groove
15–20	M18 (p. 63)	Syncopation and rests
20–25	M19 (p. 64)	Latin two-hit cell
25–30	M20 (p. 65)	Latin turnaround

Program 9 - salsa and Latin bass

Time	Block	Purpose
0–5	P5 (p. 30)	Clean string crossings
5–10	P8 (p. 33)	Muted string skips
10–15	M8 (p. 53)	Dominant chord tones
15–20	M9 (p. 54)	Minor-seven chord tones
20–25	M19 (p. 64)	Two-hit tumbao cell
25–30	M20 (p. 65)	ii–V–I–VI turnaround

Program 10 - improvisation preparation

Time	Block	Purpose
0–5	P15 (p. 40)	One-position note map
5–10	M3 (p. 48)	Major-pentatonic vocabulary
10–15	M4 (p. 49)	Minor-pentatonic vocabulary
15–20	M14 (p. 59)	Approach from below
20–25	M16 (p. 61)	Enclosures
25–30	Song drill	Use the improvisation workflow on one song

30 Improvisation quick guide

Bass improvisation is not continuous scale playing. The harmony tells you which notes are stable; rhythm determines whether those notes function as a bassline. Begin with chord tones, use scale tones to connect them, and use chromatic notes only when their destination is clear.

30.1 The note hierarchy

Priority	Note type	Use
1	Root	Establish the chord, especially at a change or phrase beginning.
2	Fifth	Stable support; useful when the chord quality is already clear.
3	Third	States major versus minor. Target it when the harmony needs to be identifiable.
4	Seventh	States major-seven, dominant-seven or minor-seven colour and often leads smoothly to the next chord.
5	Other scale tones	Connect chord tones and form melodies without leaving the key.
6	Chromatic approaches	One-fret tension immediately above or below a target chord tone.

30.2 Sixty-second preparation before playing with a song

1. Find the key or, preferably, write the chord progression.
2. Locate every chord root in one compact fretboard area.
3. Add each chord's third and fifth; add the seventh when the symbol includes one.
4. Select one scale from the table below. Treat it as a note pool, not a required sequence.
5. Choose one short rhythm or motif. Repeat it long enough that the listener can recognize it.
6. Place chord tones on important beats and let passing notes lead into them.

30.3 Fast scale chooser

Musical situation	First choice	Formula	What to listen for
Major chord or major key	Major pentatonic, p. 16	1-2-3-5-6	Safest major sound; no fourth or major seventh to rub against the chord.
Full major-key line	Major / Ionian, p. 7	1-2-3-4-5-6-7	Use 4 and 7 as movement notes unless the harmony specifically supports them.
Minor chord, uncertain context	Minor pentatonic, p. 17	1-b3-4-5-b7	Safest general minor vocabulary.
Minor chord with natural 6	Dorian, p. 9	1-2-b3-4-5-6-b7	Natural 6 creates the characteristic brighter minor sound.
Minor key with flat 6	Natural minor, p. 8	1-2-b3-4-5-b6-b7	Flat 6 is the defining darker colour.
Dominant-seven chord	Mixolydian, p. 12	1-2-3-4-5-6-b7	Major third plus flat seventh defines the dominant chord.
Blues, rock or funk dominant groove	Minor pentatonic or blues, p. 18	1-b3-4-b5-5-b7	The flat third and flat fifth are expressive tensions; resolve deliberately.
Changing chords	Chord tones first	1-3-5-7 as applicable	A correct scale does not replace following the chord changes.

30.4 When the key is unclear

1. Play only roots and fifths while listening.
2. Test the major third and minor third quietly. The chord or melody normally reveals which one belongs.
3. Listen for major seventh versus flat seventh if the chord sounds extended.
4. Remain on chord tones until the progression is understood; silence is safer than an unsupported scale run.

30.5 Five reliable ways to create a phrase

Chord-tone landing	Begin anywhere in the chosen scale but place a root, third, fifth or seventh on the next strong beat or chord change.
Motif	Create a two- or three-note idea and repeat it with one rhythmic or pitch change. Repetition makes improvisation sound intentional.
Approach note	Play one fret below or above the target, then resolve. Exercises M14 and M15 isolate this skill.
Enclosure	Surround the target from above and below before landing on it. Exercise M16 supplies a basic model.
Question and answer	Play a short phrase, leave space, then answer with a related phrase in another register or ending on another chord tone.

30.6 A first improvisation constraint

For one entire song, permit yourself only the root, third and fifth of each chord plus one chromatic approach note. Use no scale runs. This forces the improvisation to follow harmony and rhythm instead of relying on memorized shapes.

30.7 Four-stage play-along method

1. **Roots only:** establish form and chord timing.
2. **Root and fifth:** create a stable line without changing the form.
3. **Add thirds and sevenths:** make chord quality and voice leading audible.
4. **Add passing tones:** use the selected scale and chromatic approaches between established targets.

30.8 Improvising in salsa and Latin music

Rhythm first	Preserve the tumbao and the relationship to clave. A harmonically clever line that disrupts the rhythmic cycle is not an improvement.
Anticipation	Beat 4 or 4-and often points into the following measure. Know which chord the anticipated note belongs to.
Fifth to root	Moving from the fifth on 2-and toward the root on beat 4 is a practical starting cell; see M19 and M20.
Chord quality	Minor-seven and dominant-seven chords are frequent. Their thirds and flat sevenths distinguish them more clearly than a generic minor-pentatonic shape.
Chromatic motion	Approach the next root or chord tone by one fret when the rhythmic placement and destination are clear.
Restraint	Repeat a strong two-bar line. Variation should preserve the groove's identity rather than replace it every measure.

Before the track starts: identify the progression, locate roots, choose chord tones, select one scale, decide where the phrase breathes, and begin more simply than you think necessary.

31 Practice log

[illegible]

Reference principle: chord tones define the harmony; scales organize the surrounding notes; rhythm turns them into a bassline.